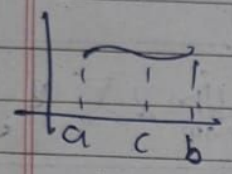
Finite Integration

① $\int_a^b x\,dx = \int_a^b t\,dt$

② $\int_a^b f(x)\,dx = -\int_b^a f(x)\,dx$

③ $\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx$



Kings rule

④ $\int_a^b f(x)\,dx = \int_a^b f(a+b-x)\,dx$

$x = (a+b-x)$

Where to apply kings rule

$$I = \int_a^b \frac{f(x)\,dx}{f(x)+f(a+b-x)}$$

↳ here answer will be ⇒ $\frac{b-a}{2}$

$$I = \int_a^b x\,\phi(x)\,dx$$

odd function = $f(-x) = -f(x)$

even function = $f(-x) = f(x)$

common example of odd function

1. $f(x) = \ln\left(\frac{a-x}{a+x}\right)$

2. $f(x) = \sqrt{1+x+x^2} - \sqrt{1-x+x^2}$

$\int_{-a}^{a} f(x)\,dx = \begin{cases} 0 & \text{if it is an odd function} \\ 2\int_{-a}^{a} f(x)\,dx & \text{if even function} \end{cases}$

$\int_{-a}^{a} \frac{dx}{1+K^{g(x)}}$ $g(x) \to$ odd

then $I = a$

$I = \int_{-a}^{a} \frac{f(x)\,dx}{1+K^{g(x)}}$ $f(x) \to$ even, $g(x) \to$ odd

then $I = \int_{-a}^{a} f(x)\,dx$

$\int_0^{\frac{\pi}{2}} \sin^2 x\,dx = \int_0^{\pi/2} \cos^2 x\,dx = \frac{\pi}{4}$

Integration of Modulus

* solve by breaking modulus or solve by making graph and finding area
* if there is one normal and other in modulus then solve by putting $|f(x)|$ +ve & −ve then break and then integrate

Integration by Periodic function

$$\int_0^{nT} f(x)\,dx = n\int_0^{T} f(x)\,dx$$

$$\int_a^{a+T} f(x)\,dx = \int_0^{T} f(x)\,dx$$

$$\int_{mT}^{nT} f(x)\,dx = (n-m)\int_0^{T} f(x)\,dx$$

$$\int_0^{\pi} |\sin x|\,dx = \int_0^{\pi} |\cos x|\,dx = 2$$

Integration of Greatest Integer function

$$\int_0^n [x]\,dx = \frac{(n-1)n}{2}$$

$$\int_0^n \{x\}\,dx = \frac{n}{2}$$

$$\frac{\int_0^n [x]\,dx}{\int_0^n \{x\}\,dx} = (n-1)$$

if limits are decimal

take the ~~the~~ thing in GIF as $f(x)$

~~ea~~ now put in value of the limit in $f(x)$ the write all the ~~numbers~~ integers ~~a~~ between f(lowest limit) & f(highest limit) equate to $f(x)$ then find all the intervals and integrate b/w them

$I_1 = \int_0^1 (1-x^n)^k\,dx$

$I_2 = \int_0^1 (1-x^n)^{k+1}\,dx$

then $\frac{I_1}{I_2} = \frac{1+n(k+1)}{n(k+1)}$

$\int_0^{2a} f(x)\,dx = \begin{cases} 2\int_0^a f(x)\,dx & f(2a-x) = f(x) \\ 0 & f(2a-x) = -f(x) \end{cases}$

$\int_a^b f(x)\,dx + \int_{f(a)}^{f(b)} f^{-1}(x)\,dx = bf(b) - af(a)$

Limit as Sum

in question take $\left(\frac{1}{n}\right)$ common somehow then write as

$$\lim_{n\to\infty} \frac{1}{n} \sum_{r=1}^{n} \frac{r}{n}$$ → nth term

take $\frac{r}{n} \to x$

$\frac{1}{n} \to dx$ | $\lim_{n\to\infty}$ & Σ as $\int$

and then integrate

Lebnitz theorem

$$I'(x) = \int_{\psi(x)}^{\phi(x)} f(t)\,dt$$

$$\Rightarrow I'(x) = f(\psi(x))\,\psi'(x) - f(\phi(x))\,\phi'(x)$$

Indefinite Integration

① $\int \frac{1}{x} dx = \ln x + c$

② $\int \frac{dx}{ax+b} = \frac{\ln(ax+b)}{a} + c$

③ $\int \sin x \, dx = -\cos x + c$

④ $\int \cos x \, dx = \sin x + c$

⑤ $\int \tan x \, dx = \ln(\sec x) + c$

⑥ $\int \cot x \, dx = \ln(\sin x) + c$

⑦ $\int \operatorname{cosec} x \, dx = \ln(\operatorname{cosec} x \cot x) + c$

⑧ $\int \sec x \, dx = \ln(\sec x + \tan x) + c$

⑨ $\int \sec^2 x \, dx = \tan x + c$

⑩ $\int \operatorname{cosec}^2 x \, dx = -\cot x + c$

⑪ $\int \sec x \tan x = \sec x + c$

⑫ $\int \operatorname{cosec} x \cot x \, dx = -\operatorname{cosec} x + c$

Substitutions

$\sin^2 x = \frac{1 - \cos(2\theta)}{2}$

$\cos^2 x = \frac{1 + \cos(2\theta)}{2}$

$\sin 3\theta = 3\sin\theta - 4\sin^3\theta$

$\cos 3\theta = 4\cos^3\theta - 3\cos\theta$

SOME SPECIAL INTEGRAL

$I = \int \sin^m x \cdot \cos^n x \, dx$

Case ①

ⓐ m, n either even/odd
=> which ever even power take as (t)

ⓑ m & n both odd
=> take any (t)

ⓒ m & n both even
=> convert to trignometric identities

Case ②

$\frac{m+n}{2} - 1 < 0$ = put cot x or tan x as (t)

INTEGRATION BY PARTIAL FUNCTION

TYPE - ① = $\int \frac{dx}{x(x+1)}$

Write as $\frac{A}{x} + \frac{B}{(x+1)}$

② ~~Find~~ compare coeff

③ Integrate (Find A & B)

TYPE - ② $\int \frac{dx}{(t^2)(t+1)}$

Write like this for powers

$\frac{A}{t} + \frac{B}{t^2} + \frac{C}{t+1}$ | only when no repeating unit

Trick if its like $\frac{A}{x-1} + \frac{B}{(x+2)} + \frac{C}{(x+3)}$

then to find A Find x for denominator $x-1=0 \therefore x=1$

then put it in other denominators like $\frac{1}{(1+2)(1+3)} = \frac{1}{(3)(4)} = \frac{1}{12}$

Do for all

TYPE ③ → if denominator has quadratic

$\frac{2x+7}{(x+1)(x^2+4)} = \frac{A}{(x+1)} + \frac{Bx+C}{(x^2+4)}$ → upper mann se

Note when linear / quadratic dx then →

Integration method

① Whenever $\ln(x)$ & x in same function and x is in denominator $[\ln(x) = t]$

Integration by parts

① $\int I\,II\,dx = I\int II\,dx - \left(\int\left(\frac{d(I)}{dx}\right)\int II\,dx\right)dx$

② $\int e^x(f(x) + f'(x))\,dx \Rightarrow e^x f(x) + C$

IMPORTANT FORMULAS

① $\int \frac{dx}{x^2+a^2} = \frac{1}{a}\tan^{-1}\left(\frac{x}{a}\right) + C$

② $\int \frac{dx}{x^2-a^2} = \frac{1}{2a}\ln\left|\frac{x-a}{x+a}\right| + C$

③ $\int \frac{dx}{a^2-x^2} = \frac{1}{2a}\ln\left|\frac{a+x}{a-x}\right| + C$

① $\int \frac{dx}{\sqrt{a^2-x^2}} = \sin^{-1}\left(\frac{x}{a}\right) + C$

② $\int \frac{dx}{\sqrt{a^2+x^2}} = \ln\left|x + \sqrt{a^2+x^2}\right| + C$

③ $\int \frac{dx}{\sqrt{x^2-a^2}} = \ln\left|x + \sqrt{x^2-a^2}\right| + C$

Substitutions (contd)

$\sqrt{\frac{a-x}{a+x}}$ or $\sqrt{\frac{a+x}{a-x}}$

$\therefore x = a\cos 2\theta$

$\sqrt{\frac{x-\alpha}{\beta-x}}$ or $\sqrt{(x-\alpha)(\beta-x)}$

$\therefore x = \alpha\cos^2\theta + \beta\sin^2\theta$

When $\sin x$ & $\cos x$ are together ~~multiplied~~ the divide the whole functions numerator and denominator with $\cos x$'s highest power

① $\int \sqrt{a^2-x^2}\,dx = \frac{1}{2}x\sqrt{a^2-x^2} + \frac{a^2}{2}\sin^{-1}\left(\frac{x}{a}\right) + C$

② $\int \sqrt{a^2+x^2}\,dx = \frac{x}{2}\sqrt{a^2+x^2} + \frac{a^2}{2}\ln\left|x + \sqrt{a^2+x^2}\right| + C$

③ $\int \sqrt{x^2-a^2}\,dx = \frac{x}{2}\sqrt{x^2-a^2} + \frac{a^2}{2}\ln\left(x + \sqrt{x^2-a^2}\right) + C$

SUBSTITUTIONS

$a^2 + x^2 = x$ t $x = a\tan\theta$ or $a\cot\theta$

$\sqrt{a^2+x^2} = x = a\sin\theta$

$a^2 - x^2 = a\sin\theta$ or $a\cos\theta$

$x^2 - a^2 = a\sec\theta$ or $a\cos\theta$

$\sqrt{x^2-a^2}$ $x = a\sec\theta$